AN EXTENSION OBTAINED BY ADJOINING A ROOT OF THE PREPARED POLYNOMIAL IS GALOIS AND FALLS INTO ONE OF FOUR CASES ACCORDING TO THE VALUATION OF THE PARAMETER

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Allowed standard prerequisites (editorial).

- Kummer theory in the presence of a primitive p-th root of unity. invoked at proof lines 35045-35048. Gives that the extension is Galois of order dividing p, which makes the four-case analysis exhaustive.
- The preparation lemma's polynomial P and its addition formula. used at statement line 35020 and proof lines 35045-35046. Supplies the polynomial whose root is adjoined and the rewriting of that adjunction.
- Ramification theory for extensions of a discrete valuation ring, including the multiplicativity of ramification indices and the description of weakly unramified extension. used at proof lines 35049-35051 and 35058-35071. Is the framework in which the four cases are separated and the implications (A), (B) and (C) are established.

Source reference. Original lemma 09I6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12029–12029. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 34977: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34977–34977. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original section 33172: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33172–33172. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a discrete valuation ring of mixed characteristic $(0, p)$ which contains a primitive pth root of 1. Let $P(t) \in A[t]$ be the polynomial of Lemma 34977. Let $\xi \in K$. Let L be an extension of K obtained by adjoining a root of $P(z) = \xi$. Then L/K is Galois and one of the following happens*

(1) *$L = K$,*
(2) *L/K is unramified with respect to A of degree p,*
(3) *L/K is totally ramified with respect to A with ramification index p, and*
(4) *the integral closure B of A in L is a discrete valuation ring, $A \subset B$ is weakly unramified, and $A \to B$ induces a purely inseparable residue field extension of degree p.*

Let π be a uniformizer of A. We have the following implications:

(A) *If $\xi \in A$, then we are in case (1) or (2).*
(B) *If $\xi = \pi^{-n}a$ where $n > 0$ is not divisible by p and a is a unit in A, then we are in case (3)*
(C) *If $\xi = \pi^{-n}a$ where $n > 0$ is divisible by p and the image of a in κ_A is not a pth power, then we are in case (4).*

Proof. Adjoining a root of $P(z) = \xi$ is the same thing as adjoining a root of $y^p = w^p(1 + \xi)$. Since K contains a primitive pth root of 1 the extension is Galois of order dividing p by the discussion in Fields, Section 09I6. It immediately follows from the discussion in Section 33172 that we are in one of the cases (1) – (4) listed in the lemma.

Case (A). Here we see that $A \to A[x]/(P(x) - \xi)$ is a finite étale ring extension. Hence we are in cases (1) or (2).

Case (B). Write $\xi = \pi^{-n}a$ where p does not divide n. Let $B \subset L$ be the integral closure of A in L. If $C = B_{\mathfrak{m}}$ for some maximal ideal $\mathfrak{m}$, then it is clear that $p\text{ord}_C(z) = -n\text{ord}_C(\pi)$. In particular $A \subset C$ has ramification index divisible by p. It follows that it is p and that $B = C$.

Case (C). Set $k = n/p$. Then we can rewrite the equation as

$$(\pi^k z)^p - \pi^{n-k}(\pi^k z) + \sum a_i \pi^{n-ik}(\pi^k z)^i = a$$

Since $A[y]/(y^p - \pi^{n-k}y - \sum a_i \pi^{n-ik} y^i - a)$ is a discrete valuation ring weakly unramified over A, the lemma follows. □